

Mohamed Emam

AI Engineer | LLM/VLM Systems, Multi-Agent Architecture & Generative AI

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SUMMARY

AI Engineer specializing in production-grade Agentic AI, high-throughput LLM/VLM serving, and real-time multimodal architectures. Experienced in architecting end-to-end distributed AI platforms—from iterative multi-agent pipelines (Langchain, LangGraph, Google-adk, openai-sdl), low-latency retrieval (RAG, Rust, BigQuery Vector), and speech-to-avatar generation (Flow Matching, wav2vec2) to sovereign inference clusters (vLLM, SGLang, llama.cpp). Active open-source contributor across top-tier agentic, memory, and generative AI codebases.

PROFESSIONAL EXPERIENCE

AI Engineer

COA Holding Company

Jan 2026 – Oct 2026

Cairo, Egypt (Hybrid)

- **Echolyzer (Call Center QA):** Architected an AI evaluation platform ingesting Bitrix24 audio calls; fine-tuned open-source LLMs to automate compliance auditing, agent scoring, and anomaly detection with real-time BI dashboard sync.
- **Auto-Camp (Campaign Optimization):** Engineered autonomous ad-management agents executing continuous KPI monitoring, dynamic budget allocation, and algorithmic ad-targeting optimization workflows.
- **Omni-Meta (Enterprise Conversational AI):** Built an omnichannel conversational platform across messaging services handling multi-turn tenant data retrieval, appointment booking, and CRM pipeline automation.

AI Engineer

TechTroll

Aug 2025 – Dec 2025

New Cairo, Egypt (Remote)

- **CairoAI (Regulatory Intelligence):** Architected an end-to-end document intelligence platform assessing Saudi banking regulations, executing clause-level compliance gap analysis and generating audit-ready reports.
- **Document Ingestion & Parsing Pipeline:** Built a high-throughput FastAPI parsing pipeline supporting PDF, DOCX, PPTX, scans, and tables using OCR, semantic chunking, and token-aware context truncation to eliminate context loss.
- **Sovereign LLM/VLM Inference Gateway:** Unified vLLM, llama.cpp, and SGLang into a local inference routing cluster for financial data sovereignty; authored a CUDA process manager with an LRU model-cache eviction mechanism to prevent VRAM/disk exhaustion.

AI Engineer

VRTualize Srl

Nov 2024 – Jul 2025

Milan, Italy (Remote)

- **Maika (Multi-Agent Travel Platform):** Spearheaded a LangGraph multi-agent system combining RAG, vector search, and dynamic web crawlers for real-time flight, hotel, and destination discovery.
- **Rust Retrieval Infrastructure:** Engineered a high-concurrency Rust-based search tool, slashing agent vector query latency to sub-3 seconds and reducing end-to-end multi-agent execution times to under 10 seconds.
- **Healthcare Predictive Analytics:** Developed longitudinal time-series forecasting pipelines for diabetes and cardiovascular disease indicators, enabling automated patient trend monitoring and risk forecasting.

AI Engineer

Deeb Realities

Apr 2024 – Nov 2024

Cairo, Egypt (Hybrid)

- **Ava (Generative Talking Avatar):** Implemented a Flow Matching generative architecture for photorealistic portrait animation from speech with wav2vec2 alignment, accelerating inference by 50%+ over diffusion models with 7 controllable emotional states.
- **Enterprise Conversational Platform:** Sole AI engineer who designed and deployed a WhatsApp, Instagram, and Messenger platform via LangGraph, FastAPI, and BigQuery Vector RAG, synced with Odoo CRM at 99.9% uptime on GCP.
- **HIM (Real-Time Voice & Chat Backend):** Developed WebSocket streaming chat infrastructure featuring parallel/sequential multi-model feedback loops, message persistence, and semantic discovery search.

KEY PROJECTS

- **llmog - VLM Object Grounding & Auto-Annotation Framework** ([GitHub](#) | Python, vLLM, llama.cpp, Pydantic): Built an open-source framework implementing an iterative Detector-Judge loop for zero-shot visual grounding; features sub-patch image tiling with NMS for small-object detection, Set-of-Mark (SoM) prompting, crop-verify validation, automated YOLO dataset relabeling, and native process lifecycle managers for local vLLM and llama-server instances.
- **MeshForge 3D - Generative Neural Mesh Synthesis Engine** ([GitHub](#) | PyTorch, CUDA, C++, Hunyuan3D-2, Docker): Engineered an end-to-end generative system converting 2D images into textured .glb meshes; compiled custom C++/CUDA rasterizers ([hy3dgen](#)) from source for hardware-accelerated surface rendering, built a thread-safe singleton managing multi-gigabyte diffusion weights across CUDA/CPU memory, and containerized the pipeline with uv.
- **DynaPrompt - Dynamic Prompt Orchestration Engine** ([GitHub](#) | Python, Pydantic v2): Designed a production-ready prompt management library featuring lazy loading, Jinja2 dynamic templating, schema validation via Pydantic v2, and native token-budget controls.
- **Automatic Multimodal Grading Engine** (VLM, RAG, FastAPI): Implemented an automated academic grading pipeline evaluating complex handwritten answers, achieving 94% accuracy (English) and 86% (Arabic), cutting manual grading labor by 80%.
- **VibeVoice Streaming API & SQL Agent** ([Github](#)) (FastAPI, WebSockets, LLM Agents): Engineered a low-latency multi-speaker text-to-speech engine over WebSockets combined with an autonomous SQL agent enabling natural language database querying.

OPEN SOURCE & ENGINEERING CONTRIBUTIONS

- **TMelyralab/MuseTalk** ([PR #397](#), [#398](#)): Contributed production FastAPI serving architecture, multi-platform Docker containerization (Linux & Apple Silicon), automated checkpoint download scripts, and demo notebooks for latent-space lip-sync generation.
- **karpthy/autoresearch** ([PR #621](#)): Added single-GPU execution support by architecting an automated VRAM lifecycle manager coordinating dynamic memory swapping between the autonomous LLM agent and model training routines.
- **browser-use/browser-use** ([PR #3499](#), [#3512](#), [#3513](#)): Resolved execution bugs and enhanced browser automation stability across dynamic multi-step agent web interactions.
- **BAI-LAB/MemoryOS** ([PR #22](#), [#23](#)): Contributed memory architecture refinements and bug fixes to the EMNLP 2025 Oral agent memory operating system for long-term personalized agent state retention.
- **VectifyAI/PageIndex** ([PR #101](#)) & **anomalyco/OpenCode** ([PR #33103](#)): Implemented Docling parsing support, Pydantic schema validation, and `pyproject.toml` packaging for vectorless RAG; added local LLM provider connectivity (llama.cpp, Ollama) in OpenCode.
- **mistralai/mistral-common** ([PR #139](#)) & **Lucebox-Hub** ([PR #83](#), 2.6k+ stars): Standardized API response formatting and added async streaming via `httpx`; hardened backend servers with CORS middleware, health-check probes, and defensive 503 error guards.
- **google/adk-docs** ([PR #2067](#)) & **Tryolabs/soccer-video-analytics** ([PR #13](#)): Clarified Model Context Protocol (MCP) client dependencies in Google Agent Development Kit; added dynamic label mapping to YOLOv8 video analytics.
- **Kaggle Competitions & Open Weights**: Fine-tuned open models (Falcon, Mistral, BLOOMZ); authored benchmarks for quantized llama.cpp GPU inference; developed deep Q-learning (RL) game-playing agents.

TECHNICAL SKILLS

- **Languages:** Python, C++, Rust, SQL, Bash, JavaScript/TypeScript
- **LLMs, VLMs & Multi-Agent Systems:** LangGraph, LangChain, vLLM, llama.cpp, SGLang, Agentic Workflows, Tool Calling, MCP, RAG, Object Grounding (llmog), Set-of-Mark (SoM), Prompt Optimization (DynaPrompt), Fine-tuning (Unsloth, LoRA/QLoRA)
- **Machine Learning & Generative AI:** PyTorch, 3D Mesh Generation (Hunyuan3D), Flow Matching, Lip-Sync (MuseTalk), wav2vec2, TTS/STT, Diffusion, Computer Vision (YOLO, OpenCV, VLMs), scikit-learn, ONNX, TensorRT
- **Backend & Data Infrastructure:** FastAPI, WebSockets, WebRTC, REST APIs, BigQuery Vector, PostgreSQL, Neo4j, Redis, Firestore, Docker, Linux Systems, CUDA Memory Management
- **Cloud & DevOps:** Google Cloud Platform (GCP), Azure, AWS, CI/CD pipelines, Git, Model Lifecycle Management

EDUCATION & CERTIFICATIONS

Bachelor of Computer Science and Artificial Intelligence

Banha University, Faculty of Computers and AI | **CGPA: 3.5/4.0 (Very Good)**

Sep 2020 – Aug 2024

Graduation Project: **A+**

Professional Training & Certifications: DeepLearning.AI Machine Learning Specialization | DeepLearning.AI NLP Specialization | DEPI React Developer (MCIT) | NTI Linux Administrator (70 Hours)

Languages: Arabic (Native), English (Professional Working Proficiency)